

Airbnb Global Listings

Demand Prediction, Price Analysis & Market Insights

Faculty of Engineering, CMP4011, Spring 2026, Team 1

Salma Mohamed Entsar
Abdelrahman Mostafa Gomaa
Youssef Roshdy
Mohamed Khaled Ahmed



Problem & Dataset

Business Questions: What factors influence price? Which locations/property types achieve highest occupancy? Can we predict price from listing features? What amenities correlate with high ratings? How do tourist vs. long-term rentals differ?

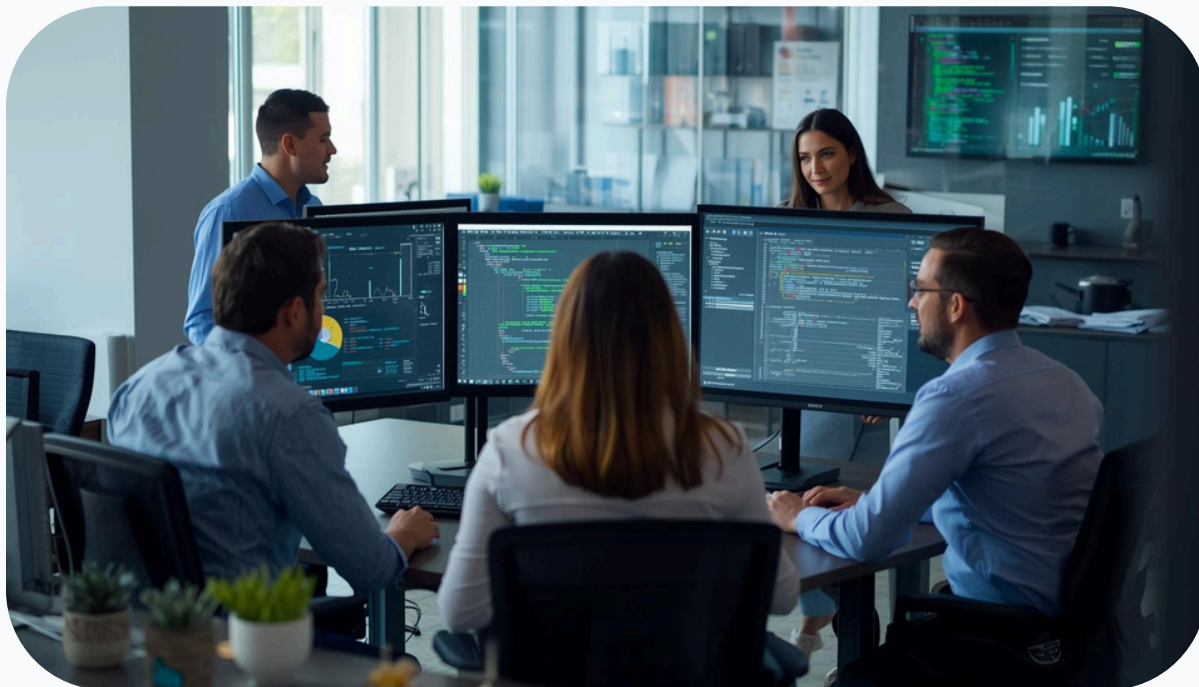
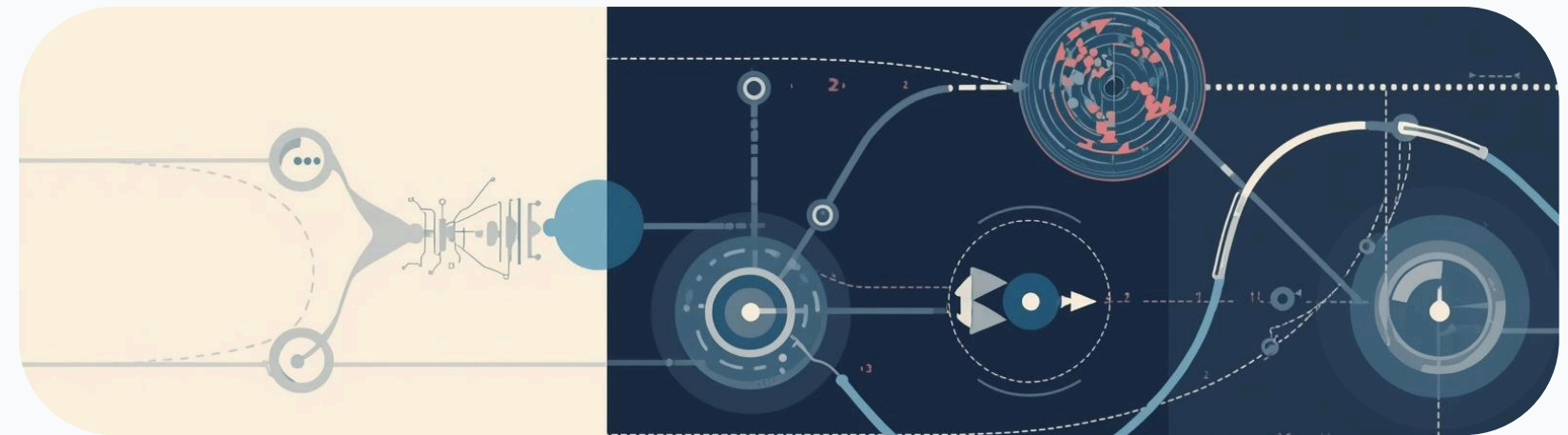


Listings: 494,954 across 100+ cities worldwide

Features: 89 features per listing

Size: 1.94 GB (Kaggle / Inside Airbnb)

End-to-End Pipeline



Leakage-free design — stats computed on training data only:

01

Raw Data (494,954 × 89) -> Preprocessing -> Label Engineering

02

EDA -> Post-processing -> K-Means Clustering -> ML Models

03

MapReduce (Hadoop) for scalable distributed computation

Results: 92.1% demand accuracy (XGBoost) | $R^2=0.83$ price regression (Random Forest) | 99.6% MapReduce overlap

Data Preprocessing

- Raw dataset: 494,954 listings × 89 features • 1.94 GB • sourced from Inside Airbnb via Kaggle
- Dropped 34 columns — URLs, free-text descriptions, scrape metadata, and high-missingness fields (Square Feet: 97.5% null, License: 97% null)
- Parsed currency columns (\$Price, Cleaning Fee), extracted binary flags from Features column, converted dates to days-elapsed features
- Amenities parsed from raw strings into a count feature + pipe-joined list for later one-hot encoding
- Removed rows with missing or zero Price -> final clean dataset: ~487,000 rows



EDA Key Findings



- Skewed distributions: Price, Min/Max Nights, and Reviews per Month are heavily right-skewed capped at 1st–99th percentile. Review Scores Rating is left-skewed (85–100), limiting discriminative power.
- Price drivers: Property size (Accommodates, Bedrooms, Bathrooms) has $r > 0.45$ with log-price; entire homes and luxury types (Villas, Boats, Boutique Hotels) command premiums.
- Geography: Top markets are NY, London, Paris, LA, Barcelona. Lat/Long alone explain ~15% of price variance — location is a primary determinant.
- Amenities: Median listing has ~20 amenities; count correlates moderately with price but not review scores. Wi-Fi/Kitchen/Heating appear in 60%+ of listings.
- Hosts: 30% of listings belong to multi-listing professionals (concentrated in NY, London, LA). Response rate is bimodal (near 100% or near 0%).
- Multicollinearity: Capacity features are intercorrelated (Accommodates VIF=13.2) -> Accommodates dropped from linear/distance-based models.

Data Postprocessing

Label Engineering

- No direct occupancy metric exists in the data — demand was proxied from booking signals
- Demand score = $0.6 \times \text{Reviews per Month} + 0.4 \times \text{normalized Number of Reviews}$
- Binary label: above training median → High demand (1), below → Low demand (0) • perfectly balanced 50/50
- 3-class label: training-set tertiles → Low / Medium / High demand

Post-processing

- 80/20 stratified train/test split • ~389,600 train • ~97,400 test • seed = 42
- All statistics computed on training data only — outlier caps, medians, encoders, scaler then applied to test
- High-cardinality categoricals → LabelEncoder • low-cardinality → one-hot • top-20 amenities → binary flags
- Accommodates dropped from linear models ($VIF = 13.2 > \text{threshold of } 10$)
- StandardScaler applied to continuous columns only — binary and one-hot columns excluded



K-Means Clustering

K=4 optimal with silhouette score 0.4469:

01 Premium/Large — High price, 3+ bedrooms, North America/Western Europe

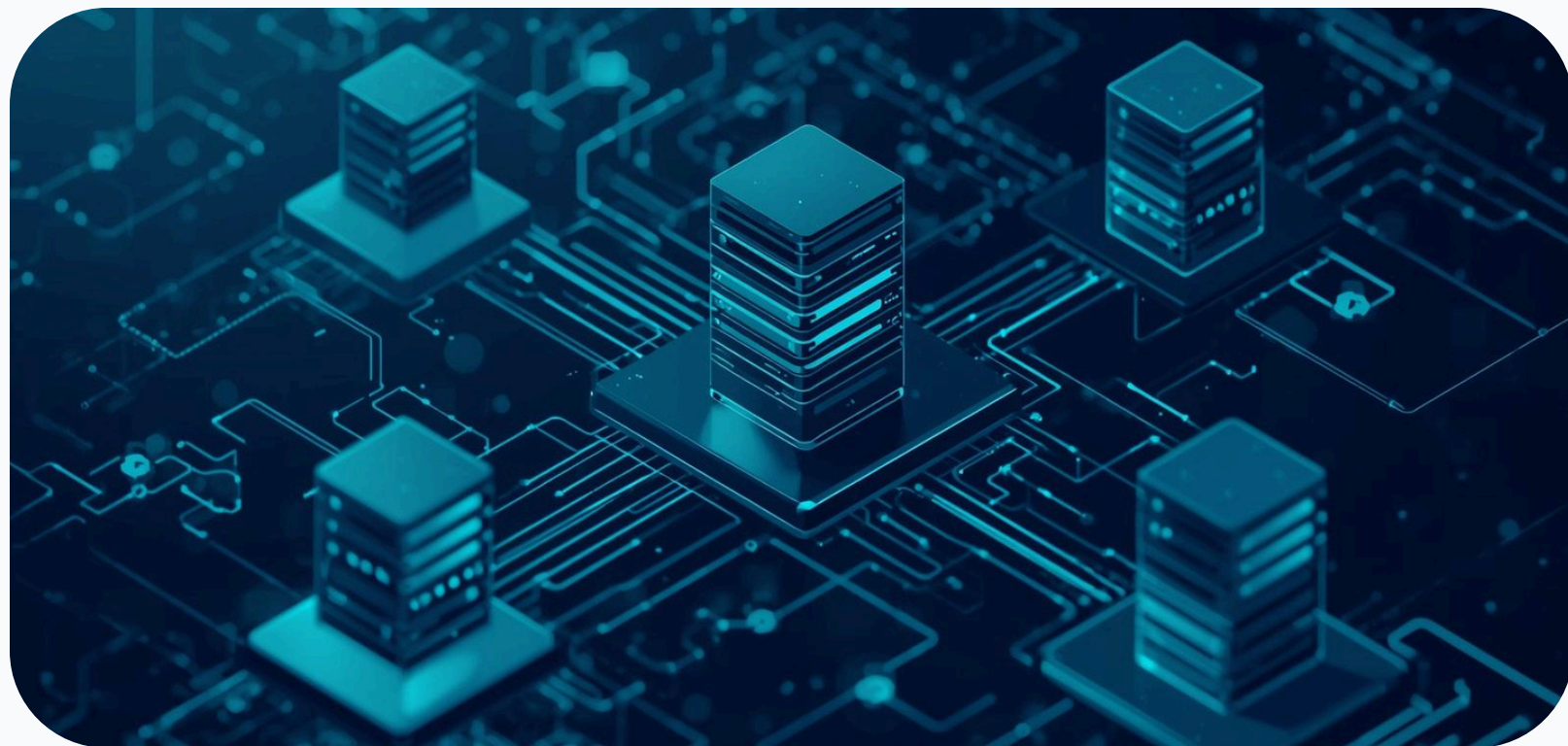
02 Mainstream — Medium price, 1 bedroom typical configuration

03 Budget/Low-rated & Southern Hemisphere clusters identified

Validation: 99.6% label overlap with scikit-learn, max centroid deviation < 0.02



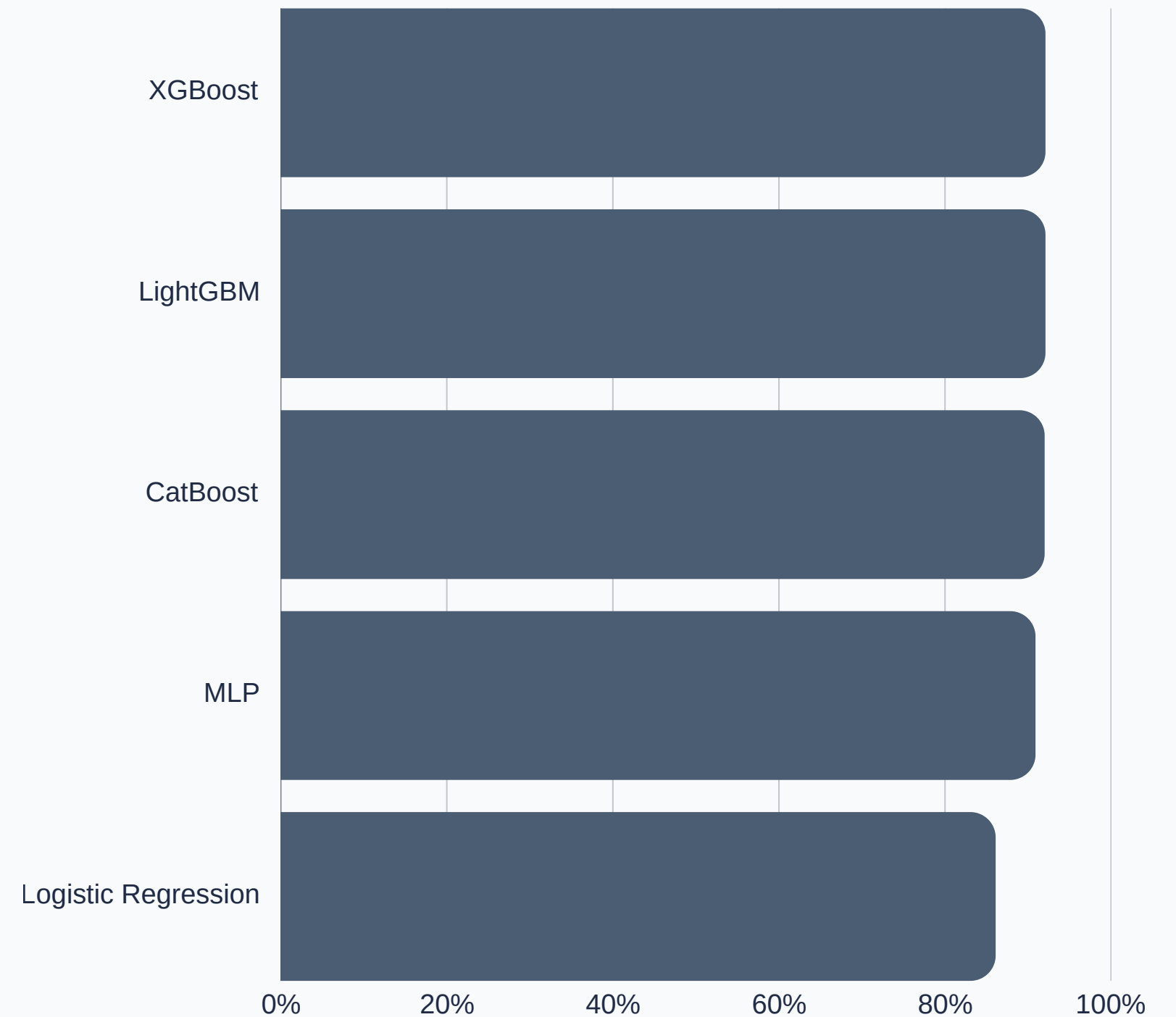
MapReduce Implementation



MapReduce enabled scalable clustering and data aggregation across 495,000+ listings with 89 features. The distributed Hadoop implementation achieved 99.6% label overlap with scikit-learn clustering results, validating accuracy at scale. This approach efficiently handled the computational complexity of processing nearly half a million records while maintaining centroid deviation below 0.02.

Demand Classification

Binary classification of demand (High vs. Low) on 97,395 test listings with balanced 50/50 classes. Boosting models dominate with 92%+ accuracy, significantly outperforming traditional methods like Logistic Regression (86.1%) and Naive Bayes (58.3%).



Source: AI-generated data. Replace or verify before use.

Price Regression

Task: Predict listing price from features using Random Forest Regression. Best performance achieved $R^2 = 0.83$, explaining 83% of price variance. Log-transformed price target was used due to heavy right-skewness in the original distribution. Top feature importance includes capacity metrics (Accommodates, Bedrooms, Beds, Bathrooms) and location features accounting for ~25% of model importance.



Source: AI-generated data. Replace or verify before use.

Results & Business Insights

For Hosts

- Adjust your price based on how recently you got reviews and how fast you respond — these are the two strongest signals our model found for predicting demand
- Your cleaning fee signals listing quality more than almost any other feature — make sure it matches your actual market tier, not just your costs
- If you're an entire home listing in NY, London, or Paris, you're in the highest-demand segment — prioritize availability and instant booking to capitalize on that
- A verified profile and near-100% response rate matter more than adding extra amenities — focus on trust before upgrades



Results & Business Insights



For Platform

- A single global model treats a New York apartment the same as a rural Thai listing — building city-level models would meaningfully improve both price prediction and demand forecasting accuracy
- The listing description, summary, and house rules were dropped from our analysis but contain rich untapped signal — applying NLP to these fields is the single highest-impact improvement available
- Our current model gives a static demand label — integrating seasonal pricing data and availability calendars would upgrade this to a real-time demand forecast that hosts can actually act on
- The demand labels were built from review counts, which is an imperfect proxy — access to verified booking records would make the entire prediction pipeline significantly more reliable
- XGBoost, LightGBM, and CatBoost each hit ~92% individually — stacking them via ensemble voting would push accuracy further without any additional data
- The current pipeline processes 1.94 GB in minutes on a single machine — migrating to AWS EMR or Azure HDInsight with Spark would enable real-time inference at scale

Thank You! Questions?

